

MEKHANA RAJESH

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PROFESSIONAL SUMMARY

Analytical and detail-oriented MCA student with a strong mathematical foundation and hands-on experience in Python, Java, Machine Learning, and Data Science. Skilled in translating complex data into actionable insights through data preprocessing, model development, and visualization. Proven ability to collaborate effectively in team environments while writing clean, efficient code. Eager to apply strong problem-solving skills and a continuous learning mindset to contribute to software development and data-driven projects in a fast-paced organization.

TECHNICAL SKILLS

Programming Languages: Python, Java, C programming
Data Science & Machine Learning: Machine Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Data Preprocessing, Data Visualization
Databases & Tools: MySQL, SQLite, ChromaDB, Git, GitHub, Jupyter Notebook, Google Colab, Microsoft Excel
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), REST APIs, Problem Solving

EDUCATION

Master of Computer Applications (MCA) 2025 – 2027 (Pursuing)
Cochin University of Science and Technology (CUSAT)
Bachelor of Science in Mathematics 2021 – 2024
Deva Matha College, Mahatma Gandhi University CGPA: 8.37

INTERNSHIP EXPERIENCE

Data Science and Machine Learning Intern — NeST Digital May 2026 – June 2026

- Performed data preprocessing and cleaning on structured datasets to improve data quality and consistency.
- Assisted in developing and testing machine learning models using Python on real-world datasets.
- Created data visualizations to identify trends and support analytical insights.
- Collaborated with the team to evaluate model performance and refine results.
- Documented workflows and methodologies to ensure clarity and reproducibility.

Generative AI Intern — Corestone June 2026

- Developed an AI-powered customer support chatbot using Retrieval-Augmented Generation (RAG) architecture.
- Built the backend using FastAPI and integrated Qwen 2.5 LLM for generating relevant and context-aware responses.
- Utilized ChromaDB for vector-based semantic search and efficient knowledge retrieval from documents.
- Managed team coordination, task distribution, and project execution to deliver the chatbot solution within the timeline.

PROJECTS

Elevora — AI Career Intelligence Platform

Tech: Python, Flask, React, NLP, PDF Generation

- Developed an AI-powered platform that analyzes resumes against job descriptions using NLP-based skill extraction.
- Implemented ATS-style resume analysis, skill-gap identification, and job readiness scoring and generated personalized career recommendations and downloadable PDF reports.
- Designed an interactive interface to help users visualize strengths and missing skills.

Budget Coach — Personal Finance Management Website

Tech: Flask, SQLite, HTML, CSS, JavaScript

- Built a full-stack personal finance application for expense tracking and budget management.
- Implemented features including expense CRUD operations, daily and monthly spending analytics, and automated budget-limit alerts to improve user financial awareness.
- Added a goal-setting and progress-tracking module along with a gamified rewards system to boost user engagement and consistency.

CERTIFICATIONS

- Data Science — Duke University (Coursera)
- Voice of Stakeholder (VoS) — Young Innovators Program 2022, Kerala Development and Innovation Strategic Council (K-DISC)
- Python Programming — Youth Employability Skill Training Co-operative Educational Society Ltd.
- Introduction to Internet of Things — NPTEL

LANGUAGES

English, Malayalam, Tamil